Name: **EXPLOREMATE – AI TOUR GUIDE**

GithubUrl: https://github.com/Devasish009/ExploreMate

Description : The AI Audio Tour Guide is one of the core features of ExploreMate that helps users explore places independently through smart audio-based guidance. Instead of relying on physical tour guides, users can listen to AI-generated descriptions, historical information, cultural insights, and travel tips while visiting a location.This feature is specially designed for solo travelers, introverts, and users who prefer self-guided exploration.

Functional Requirements

Authentication

User Registration:

- User can create a new account using:
  - Name
  - Email
  - Phone Number
  - Password
- OTP verification using Firebase Authentication

Login System:

- Secure login using Email/Username and Password
- JWT-based authentication for secure session handling
- Auto-login support until user logs out

User Roles:

- Traveler/User
- Admin

Security Features:

- Password encryption
- Token expiry and refresh mechanism
- Secure API access using JWT tokens
- Session management and logout support

User Account Management:

- Create and manage travel accounts
- Update profile and preferences

- Save favorite places and trip history

Smart Place Discovery:

Place Discovery

- Search tourist places, food spots, and hidden gems
- AI-based nearby place recommendations
- Filter by category, distance, and ratings

AI Travel Intelligence:

Smart Recommendations

- Personalized trip suggestions
- Weather-based food recommendations
- AI-generated travel tips and itineraries
- Export trip plans as PDF

Security & Authentication:

User Security

- Secure login and registration
- OTP verification using Firebase
- JWT-based authentication
- Encrypted user data protection

Smart Notifications:

- Weather and travel alerts
- Nearby attraction updates
- Mission and reward notifications

Community Features:

Gamified City Explorer

- XP points and levels
- Badges and rewards
- Leaderboard system
- Group travel and sharing features

Non-Functional Requirements

- Fast and responsive user interface

- Page loading time under 2 seconds

- API response time under 3 seconds
- Support for multiple concurrent users
- Secure communication using HTTPS/TLS
- User credentials stored securely in encrypted form
- High application availability and reliability
- Smooth performance across mobile devices
- User-friendly and accessible UI design
- Regular testing and bug fixing for stability
- Scalable architecture for future feature expansion

Problem Statement:

Travelers often face difficulties in planning trips, discovering hidden places, understanding local culture, and getting personalized travel guidance. Existing travel applications mainly focus on booking and navigation and lack intelligent, interactive, and personalized exploration features. Solo travelers and introverts especially need a smart solution that helps them explore places independently and comfortably.

Proposed Solution:

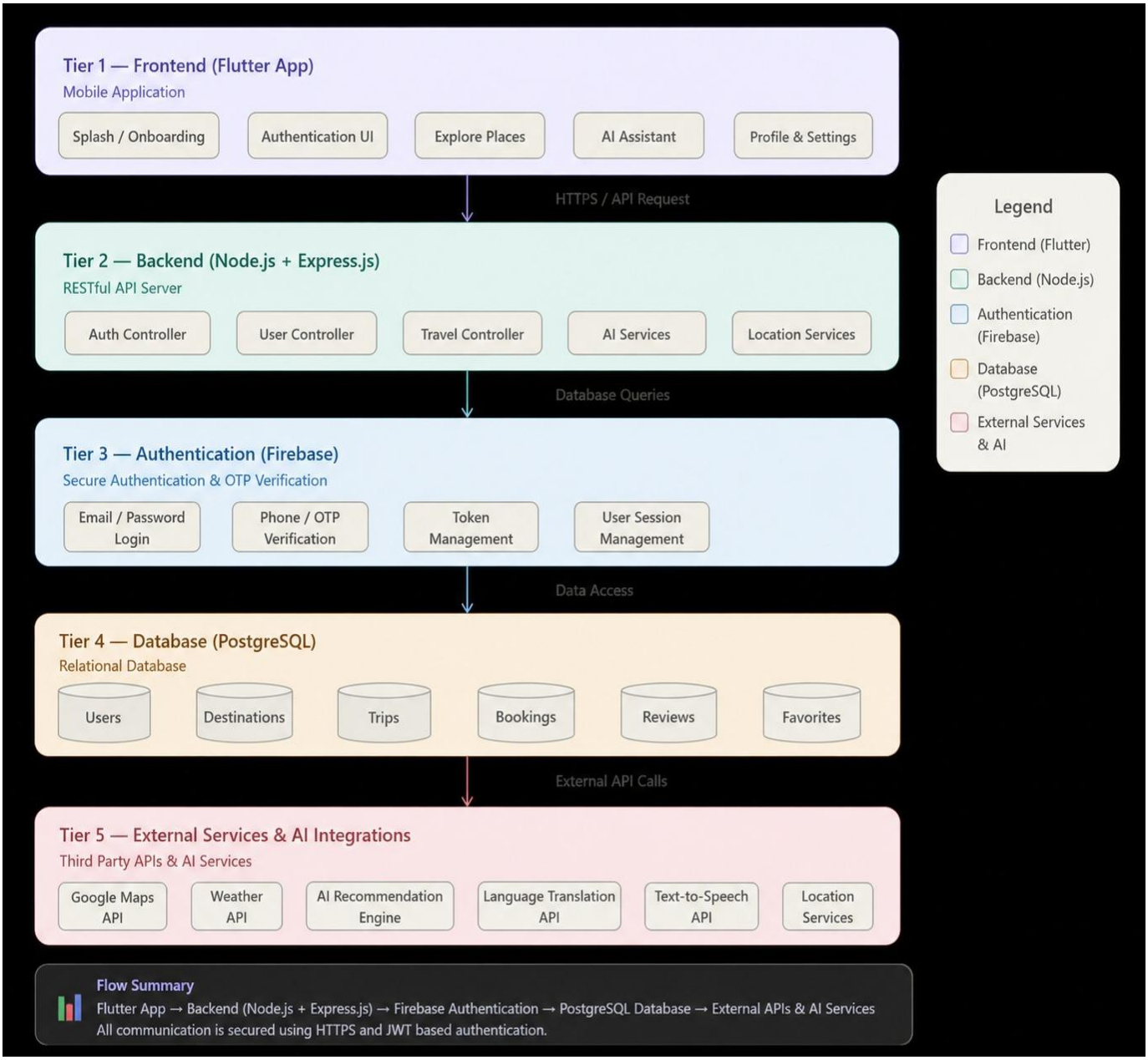
ExploreMate is an AI-powered travel companion application designed to provide smart and personalized travel experiences. The application helps users discover hidden gems, receive weather-based food suggestions, plan trips, translate languages, and explore cities using gamification features like XP points, badges, and missions. It also includes an AI Audio Tour Guide that provides audio-based place guidance and interactive travel assistance.

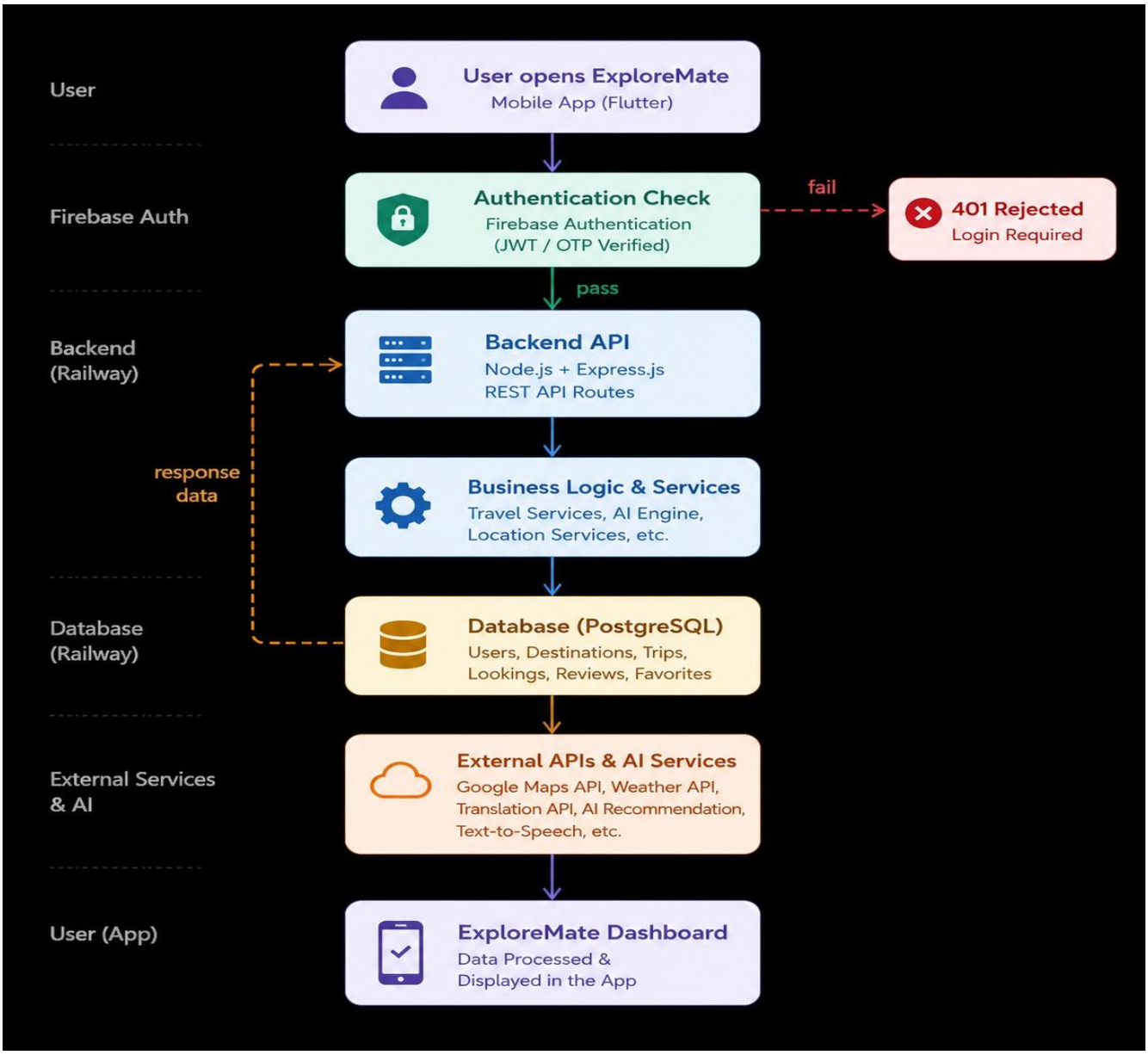
Technologies Used:

ExploreMate is developed using Flutter for frontend mobile application development and Node.js with Express.js for backend services. PostgreSQL is used as the database, while Firebase Authentication provides secure OTP-based login and registration. The application integrates APIs such as Google Maps API, Weather API, Translator API, Text-to-Speech API, and OpenAI API for intelligent travel features and AI-based content generation.

System Architecture:

The ExploreMate system follows a layered architecture where the Flutter mobile application communicates with the Node.js and Express.js backend server. Firebase Authentication handles secure user login and OTP verification, while PostgreSQL stores user and travel-related data. External APIs and AI services are integrated into the backend to provide smart recommendations, translation support, audio guidance, weather-based suggestions, and location-based services efficiently.


Tier 1 — Frontend (Flutter App)
Mobile Application
Splash / Onboarding
Authentication UI
Explore Places
AI Assistant
Profile & Settings
HTTPS / API Request
Tier 2 — Backend (Node.js + Express.js)
RESTful API Server
Auth Controller
User Controller
Travel Controller
AI Services
Location Services
Database Queries
Tier 3 — Authentication (Firebase)
Secure Authentication & OTP Verification
Email / Password Login
Phone / OTP Verification
Token Management
User Session Management
Data Access
Tier 4 — Database (PostgreSQL)
Relational Database
Users
Destinations
Trips
Bookings
Reviews
Favorites
External API Calls
Tier 5 — External Services & AI Integrations
Third Party APIs & AI Services
Google Maps API
Weather API
AI Recommendation Engine
Language Translation API
Text-to-Speech API
Location Services
Legend
Frontend (Flutter)
Backend (Node.js)
Authentication (Firebase)
Database (PostgreSQL)
External Services & AI
Flow Summary
Flutter App → Backend (Node.js + Express.js) → Firebase Authentication → PostgreSQL Database → External APIs & AI Services
All communication is secured using HTTPS and JWT based authentication.



In Scope:

The ExploreMate application includes AI-powered travel assistance, hidden place discovery, smart trip scheduling, weather-based food recommendations, AI Audio Tour Guide, translator support, gamified city exploration, secure authentication, personalized travel suggestions, travel activity tracking, notifications, and downloadable travel reports.

Out Scope:

The current version of ExploreMate does not include hotel or flight booking systems, offline map navigation, AR/VR travel experience, wearable device integration, real-time multiplayer gaming features, advanced AI prediction systems, or complete desktop web application support.

Future Enhancements

Future enhancements of ExploreMate include dark mode support, offline audio guide downloads, multi-language voice assistance, real-time navigation guidance, community travel sharing, emergency support features, AI-based personalized storytelling, and AR/VR-based virtual travel exploration.

Conclusion:

ExploreMate is an AI-powered travel companion application that combines intelligent guidance, personalization, and gamification to improve the travel experience. The application helps users explore places independently through smart recommendations, audio-based guidance, hidden place discovery, and interactive travel features. With modern technologies and scalable architecture, ExploreMate provides a strong foundation for future travel innovation.

Project Type:

AI-Powered Mobile Travel Application